

Economic History II

Causality and colonial origins

FGV EPGE

Outline of the class

- Introduction to causality.
- The potential outcomes model.
- AJR (2001): colonial origins and comparative development.

The causal question

- What is the effect of receiving a treatment D on an outcome Y ?
- Examples: aspirin and headaches; training and wages; class size and learning.
- The word “effect” compares what happened with what would have happened under a different choice.

Two potential outcomes

With treatment

$$Y_i(1)$$

Headache one hour after taking aspirin.

Without treatment

$$Y_i(0)$$

Headache one hour after not taking aspirin.

$$\tau_i = Y_i(1) - Y_i(0)$$

individual causal effect

The observed outcome

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0), \quad D_i \in \{0, 1\}$$

- If $D_i = 1$, we observe $Y_i(1)$; $Y_i(0)$ is missing.
- If $D_i = 0$, we observe $Y_i(0)$; $Y_i(1)$ is missing.
- The observed data mix treatment assignment with the realized potential outcome.

The missing counterfactual

Unit	D_i	We observe	We do not observe
Person with aspirin	1	$Y_i(1)$	$Y_i(0)$
Person without aspirin	0	$Y_i(0)$	$Y_i(1)$

The problem is the lack of data on the unobserved world.

First best: observing both worlds

- The ideal would be to observe, for the same unit and at the same moment, $Y_i(1)$ and $Y_i(0)$.
- Then τ_i would be just an observed difference.
- This is impossible: once D_i is realized, only one outcome enters the data.

Second best: randomization

- Randomization does not create the individual counterfactual.
- It makes treated and control groups comparable on average.
- The control group then tells us, on average, the outcome the treated group would have had without treatment.

What randomization solves

$$D_i \perp (Y_i(1), Y_i(0))$$

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \mathbb{E}[Y_i(1) - Y_i(0)]$$

- Observed differences reflect the treatment, not the composition of the groups.
- Statistical uncertainty and questions of external validity still remain.

Third best: observational methods

- For many questions, who receives treatment is not chosen by lottery.
- In observational studies, we need to replace randomization with an identification argument.
- Examples: controls, matching, instrumental variables, differences-in-differences, regression discontinuity.

Identification in observational studies

1. Who received the treatment, and why?
2. Which group or variation tells us the counterfactual?
3. What other changes could explain the same result?

The empirical strategy answers these three questions.

Historical variation

- In economic history, many treatments cannot be randomized.
- History sometimes generates differences in exposure across places, cohorts, sectors or periods.
- Using that variation is still observational: the question is whether it helps build a credible counterfactual.

Cantoni and Yuchtman: three objectives

1. Understanding specific historical processes: medieval universities, monasteries, the Commercial Revolution.
2. Testing economic hypotheses in historical settings: comparative advantage, agglomeration, property rights.
3. Linking variation in the past to present-day outcomes: colonial institutions, slavery, the Peruvian mita.

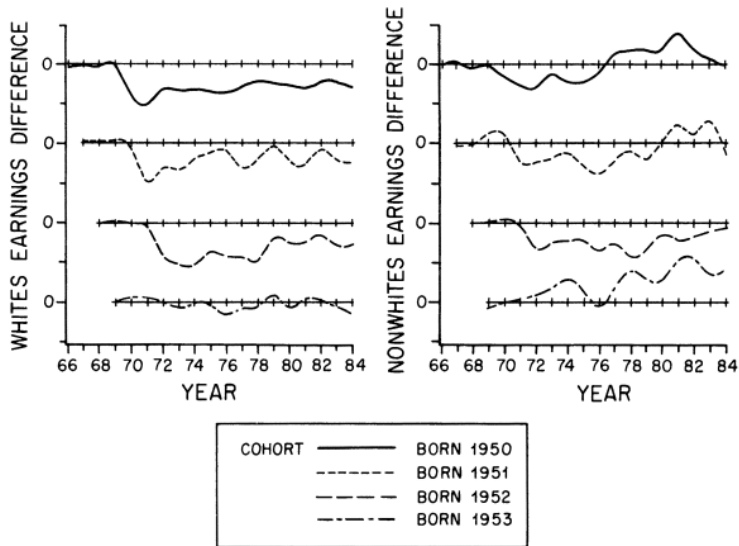
Examples of variation

Example	Question	Source of variation
Angrist (1990)	Military service and earnings	Draft lottery
Angrist and Krueger (1991)	Schooling and wages	Compulsory schooling laws and quarter of birth
Bleakley (2007)	Health and human capital	Exposure to the eradication campaign
Dell (2010)	Coerced labour and current income	Historical boundary of the Peruvian mita
AJR (2001)	Institutions and income	Settler mortality

Lifetime Earnings and the Vietnam Era Draft Lottery: Evidence from Social Security Administrative Records

By JOSHUA D. ANGRIST*

The randomly assigned risk of induction generated by the draft lottery is used to construct estimates of the effect of veteran status on civilian earnings. These estimates are not biased by the fact that certain types of men are more likely than others to service in the military. Social Security administrative records indicate that in the early 1980s, long after their service in Vietnam was ended, the earnings of white veterans were approximately 15 percent less than the earnings of comparable nonveterans. (JEL 824)



Notes: The figure plots the difference in FICA taxable earnings by draft-eligibility status for the four cohorts born 1950–53. Each tick on the vertical axis represents \$500 real (1978) dollars.

The long-run effect of compulsory military service

- Estimates the causal impact of compulsory military service on beliefs and personality traits in the long run.
- To address endogeneity concerns, it exploits the conscription lottery in Argentina.
- It combines administrative data from the lottery with data from a survey designed by the authors on beliefs and personality.
- Men who served are more likely to adopt a military mindset.
- The effect is persistent.
- Since many people go through compulsory military service, the results help us understand how beliefs and personality traits are formed.

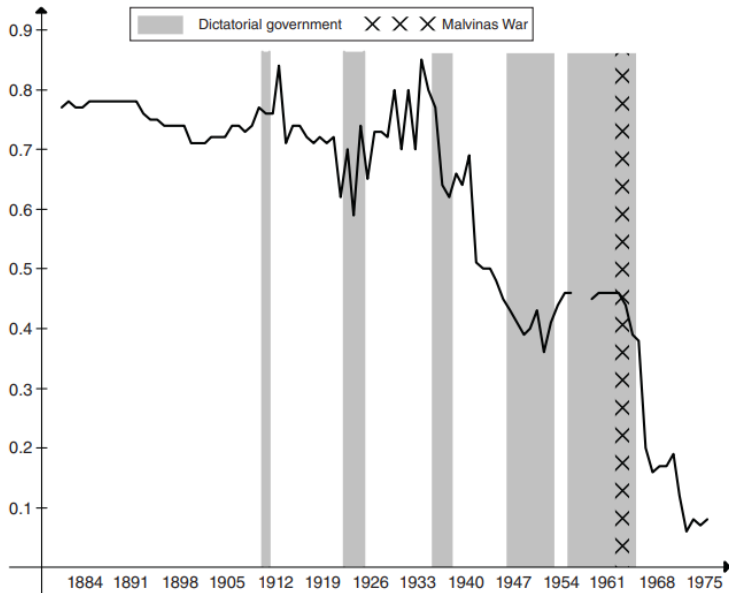


FIGURE 1. PROPORTION OF MEN SERVING IN THE ARGENTINE MILITARY, BY BIRTH COHORT

Table 1. Draft-eligibility status

Cohort	Sample size	Population proportion	Sample proportion	Difference
1958	43	0.825	0.870	-0.045
1959	70	0.680	0.681	-0.001
1960	57	0.659	0.606	0.053
1961	80	0.650	0.624	0.027
1962	64	0.680	0.735	-0.055
1963	48	0.650	0.623	0.027
1964	37	0.600	0.676	-0.076
1965	58	0.607	0.705	-0.098
1966	44	0.373	0.451	-0.078
1967	41	0.333	0.186	0.147**
1968	79	0.413	0.381	0.032
1969	68	0.446	0.526	-0.080
1970	57	0.502	0.532	-0.030
1971	81	0.281	0.264	0.017
1972	68	0.164	0.268	-0.104*
1973	57	0.240	0.203	0.037
1974	55	0.256	0.210	0.046
1975	49	0.257	0.340	-0.083
Total	1,133	0.477	0.487	-0.010

Notes: *Significant at the 10% level. **Significant at the 5% level.

***Significant at the 1% level.

Table 2. Pre-treatment characteristics, by draft-eligibility assignment

	Draft eligible mean	Non-draft eligible mean	Difference
Father's country of birth	0.920 (0.271)	0.910 (0.286)	0.010 (0.017)
Mother's country of birth	0.906 (0.292)	0.921 (0.270)	-0.015 (0.017)
His father did military conscription	0.623 (0.485)	0.632 (0.483)	-0.009 (0.029)
Father's maximum level of education			
No instruction	0.015 (0.120)	0.010 (0.101)	0.004 (0.007)
Incomplete primary school	0.129 (0.335)	0.120 (0.326)	0.009 (0.020)
Complete primary School	0.250 (0.433)	0.225 (0.418)	0.025 (0.025)
Incomplete secondary school	0.114 (0.318)	0.114 (0.318)	0.000 (0.019)
Complete secondary School	0.158 (0.365)	0.181 (0.385)	-0.023 (0.022)
Incomplete high education	0.024 (0.152)	0.033 (0.178)	-0.009 (0.010)
Complete high education	0.063 (0.244)	0.038 (0.191)	0.026* (0.013)
Incomplete university	0.073 (0.259)	0.083 (0.276)	-0.010 (0.016)
Complete university	0.165 (0.371)	0.186 (0.389)	-0.021 (0.023)
Mother's maximum level of education			
No instruction	0.011 (0.104)	0.016 (0.124)	-0.005 (0.007)
Incomplete primary school	0.116 (0.320)	0.103 (0.305)	0.013 (0.019)
Complete primary School	0.310 (0.463)	0.248 (0.432)	0.062** (0.027)
Incomplete secondary school	0.101 (0.307)	0.115 (0.370)	-0.014 (0.019)

Table A1. Survey

Survey Questions	
Personal Information	Questions
	Month of birth
	Year of birth
	District of origin
	Father's country of birth
	Father's maximum level of education
	Mother's country of birth
	Mother's maximum level of education
	Did your father do military conscription?
	Did you do military conscription?
Beliefs/Personality traits	Questions
Beliefs	Having a weapon should be a right
	The use of violence is justified to resolve certain conflicts
	Intervention from one country to another is justified under certain circumstances
	Military service should be mandatory
	A coup is acceptable when a government is incompetent

Conservatism

I tend to vote for conservative political candidates
I believe in one true religion
I believe that we should be tough on crime
I tend to vote for liberal political candidates
I believe in the importance of art
I don't consider myself religious
I believe that there is no absolute right and wrong
I believe that criminals should receive help rather than punishment

Discipline

I believe laws should be strictly enforced
I use swear words
I try to follow the rules
I oppose authority
I respect authority
I know how to get around the rules

Conscription and Crime: Evidence from the Argentine Draft Lottery[†]

By SEBASTIAN GALIANI, MARTÍN A. ROSSI, AND ERNESTO SCHARGRODSKY*

We estimate the causal effect of mandatory participation in military service on individuals' subsequent involvement in criminal activities. To identify this causal effect, we exploit the random assignment of young men to conscription in Argentina through a draft lottery. Using a dataset that includes draft eligibility, participation in military service, and criminal records, we find that conscription increases the likelihood of developing a criminal record. The effects are significant not only for cohorts that provided military service during wartime, but also for those that served during peacetime. Our results do not support the introduction of conscription for anti-crime purposes.
JEL (H56, K42, O17)

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AJR (2001)	Institutions and income	Settler mortality

DOES COMPULSORY SCHOOL ATTENDANCE AFFECT SCHOOLING AND EARNINGS?*

JOSHUA D. ANGRIST AND ALAN B. KRUEGER

We establish that season of birth is related to educational attainment because of school start age policy and compulsory school attendance laws. Individuals born in the beginning of the year start school at an older age, and can therefore drop out after completing less schooling than individuals born near the end of the year. Roughly 25 percent of potential dropouts remain in school because of compulsory schooling laws. We estimate the impact of compulsory schooling on earnings by using quarter of birth as an instrument for education. The instrumental variables estimate of the return to education is close to the ordinary least squares estimate, suggesting that there is little bias in conventional estimates.

Every developed country in the world has a compulsory schooling requirement, yet little is known about the effect these laws have on educational attainment and earnings.¹ This paper exploits an unusual natural experiment to estimate the impact of compulsory schooling laws in the United States. The experiment stems from the fact that children born in different months of the year start school at different ages, while compulsory schooling laws generally require students to remain in school until their sixteenth or seventeenth birthday. In effect, the interaction of school-entry

Malaria Eradication in the Americas: A Retrospective Analysis of Childhood Exposure[†]

By HOYT BLEAKLEY*

This study uses the malaria-eradication campaigns in the United States (circa 1920) and in Brazil, Colombia, and Mexico (circa 1955) to measure how much childhood exposure to malaria depresses labor productivity. The campaigns began because of advances in health technology, which mitigates concerns about reverse causality. Malarious areas saw large drops in the disease thereafter. Relative to non-malarious areas, cohorts born after eradication had higher income as adults than the preceding generation. These cross-cohort changes coincided with childhood exposure to the campaigns rather than to pre-existing trends. Estimates suggest a substantial, though not predominant, role for malaria in explaining cross-region differences in income. (JEL I12, I18, J13, O15)

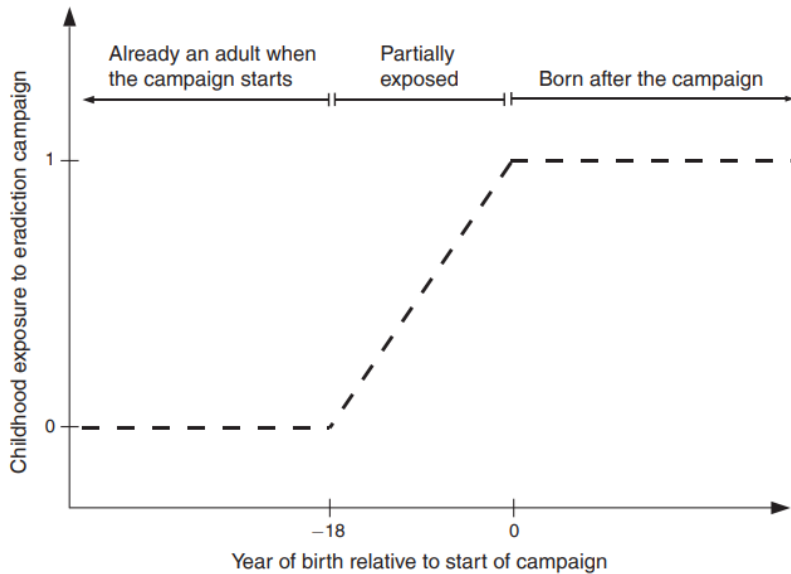


FIGURE 2. CHILDHOOD EXPOSURE TO ERADICATION CAMPAIGN

The cost of the third best

- Identifying variation can push the analysis towards where data and contrast happen to exist.
- A credible estimate still requires theory and historical interpretation.
- To link past and present, we need to follow the causal chain over time.

Colonial origins

Acemoglu, Johnson and Robinson (2001)

AJR (2001): the question

- Why are some former colonies much richer than others?
- Hypothesis: institutional differences created in the colonial period persisted and affect current income.
- Empirical problem: rich countries may have better institutions for many reasons; correlation is not enough.

The hypothesis

1. Europeans adopted different colonial strategies in different places.
2. Where European mortality was high, European settlement was less viable.
3. Less settlement favoured extractive institutions.
4. Colonial institutions persisted and influenced current institutions.

(potential) settler
mortality $\Rightarrow$ settlements

$\Rightarrow$ early
institutions $\Rightarrow$ current
institutions

$\Rightarrow$ current
performance.

What they measure

Concept	Measure	Source / interpretation
Current development	Log GDP per capita in 1995	PPP, World Development Indicators
Current institutions	Protection against expropriation, 1985–1995	Political Risk Services; scale from 0 to 10
Instrument	European settler mortality	Soldiers, bishops and sailors in the seventeenth to nineteenth centuries

Empirical strategy

- Relationship of interest:

$$\log y_i = \mu + \alpha R_i + X_i' \gamma + \varepsilon_i$$

- R_i : protection against expropriation.
- Problem: R_i is endogenous.
- Proposed solution: use potential settler mortality as an instrument for R_i .

The logic of the instrument

settler mortality → European settlement → institutions → current income

- Relevance: mortality affected the viability of European settlement.
- Exclusion: historical mortality should not affect current income except through institutions.

Log GDP per capita, PPP, 1995

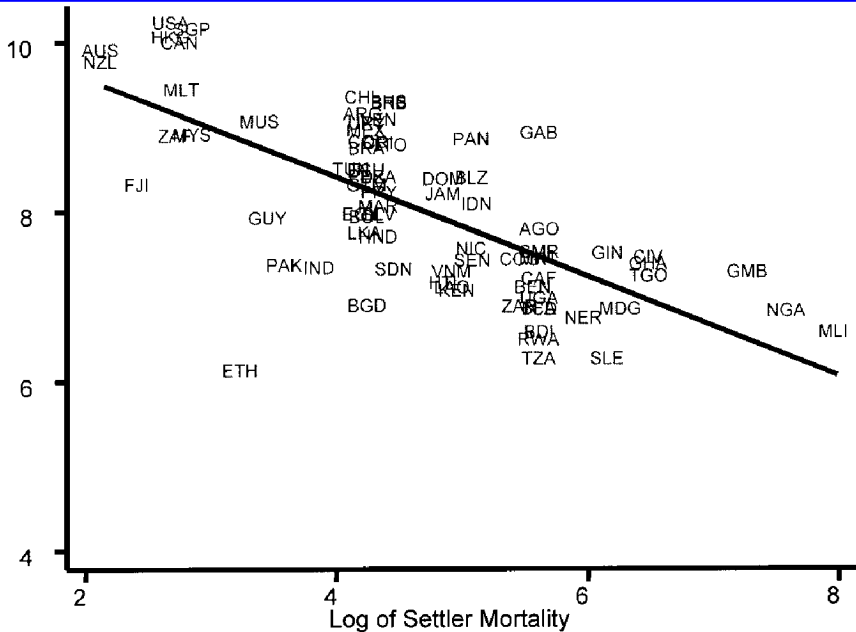


Figure 1: reduced form

- Countries where European settlers faced higher mortality tend to be poorer today.
- This is not yet the causal estimate of institutions on income.
- The figure shows the total relationship between the instrument and the outcome.

Mortality, geography and disease

- Settler mortality may be correlated with geography, disease or other persistent characteristics.
- To defend the institutional interpretation, AJR need to show the channel:

mortality $\rightarrow$ institutions $\rightarrow$ income

- The next figures separate these two parts.

Log GDP per capita, PPP, 1995

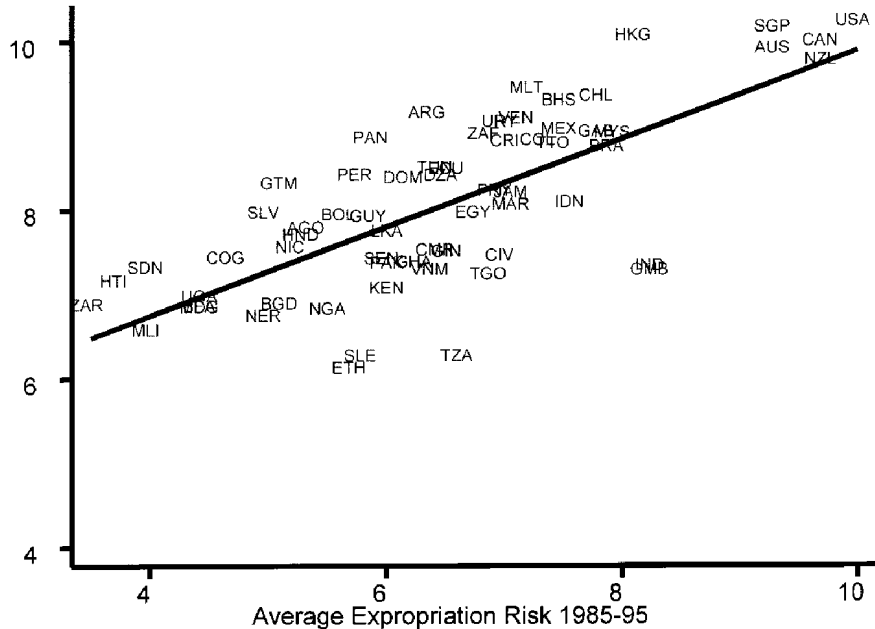


Figure 2: institutions and income

- Countries with lower expropriation risk tend to have higher income per capita.
- This is the central OLS correlation: better institutions are associated with higher income.
- But OLS may confound reverse causation, omitted variables and measurement error.

What Figure 2 leaves open

- Rich economies may be able to choose, sustain or pay for better institutions.
- The expropriation index is observed in 1985–1995, after the development process.
- The causal question requires variation in institutions that does not come from current income.

Average Expropriation Risk 1985-95

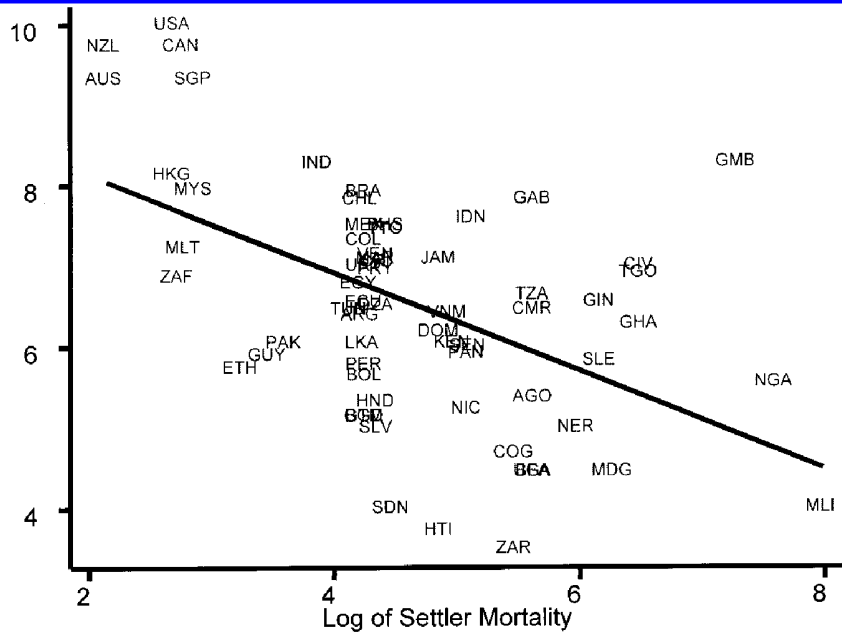


Figure 3: first stage

- Higher potential settler mortality predicts worse current institutions.
- This is the first stage of the IV: the instrument has to move the endogenous variable.
- In the paper, settler mortality explains more than a quarter of the variation in current institutions.

IV results

- AJR estimate the effect of institutions using the part of the variation explained by settler mortality.
- The benchmark IV coefficient for protection against expropriation is about 0.94.
- The institutional difference between Nigeria and Chile would imply a substantial income difference in the long run.

The exclusion restriction

- For the IV to be valid, settler mortality must affect current income only through institutions.
- Main concern: historical mortality may be picking up current disease, climate or geography.
- AJR's defence: malaria and yellow fever were far more lethal for non-immune Europeans than for local adults.

Robustness

- The results do not depend on the Neo-Europes nor on Africa alone.
- Additional controls do not eliminate the effect: latitude, climate, colonizer, legal origin, religion, natural resources, soil, disease and racial composition.
- Overidentification tests find no strong evidence of a direct effect of mortality on income.

What the paper does not settle

- Institutions enter as a broad bundle: property rights, expropriation risk, constraints on the executive, the rule of law.
- The paper does not identify exactly which institutional reform would produce the estimated gain.
- The result does not imply colonial determinism, since the colonial experience is only one of the sources of institutional variation.

Main message

- AJR propose a causal answer to the question: do economic institutions matter for development?
- The answer comes from an observational strategy: settler mortality as a source of institutional variation.
- The empirical argument rests both on the first stage and on the plausibility of the exclusion restriction.